



Republic of the Philippines

Department of Environment and Natural Resources

MINES AND GEOSCIENCES BUREAU

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LANDSLIDE AND FLOOD THREAT ADVISORIES

To: **Brgy. Chairperson
Brgy. Fortune
City of Marikina
Metropolitan Manila**

Date: Oct 9, 2024

Dear Sir/Madam:

Please be advised that the Geohazards Mapping and Assessment Team (GMAT) of the Mines and Geosciences Bureau (MGB) has conducted **landslide and flood assessment** for the updating of 1:10,000 Scale Geohazard Map in your barangay. The following are the results and recommendations following the assessment:

Purok / Sitio / Area of Concern	GPS Coordinates (Coordinate System: Luzon 1911)	Flood Susceptibility Rating	Geohazards Assessment
Marikina DRRMO	14°39'51.4" N 121°07'12.4" E	Low	The Nangka River traverses the city on its northeastern boundary, which also serves as the northern boundary of Brgy. Fortune. The overtopping of the channel during inclement weather conditions has been a major source of flooding within the barangay. Flood heights of more than 2 m were reported on residential settlements close to the said channel such as in San Miguel Phase II and III. Further, evidences of riverbank erosion were reported and observed especially on roads adjacent to the channel and/or its tributaries.
Barangay Hall	14°39'52.4" N 121°07'34.3" E	Moderate	
San Miguel Phase II	14°39'47.5" N 121°07'52.2" E	Low	
San Miguel Phase III	14°39'52.0" N 121°07'49.5" E	High	
	14°39'54.4" N 121°07'48.2" E	Very High	
Champaca II	14°39'46.1" N 121°07'35.0" E	Very High	Tributaries of the channel may also cause inundation especially when aggravated by urbanization. Localized flooding due to low elevation were also reported.
Bonanza	14°39'49.7" N 121°07'10.2" E	High	

Remarks / Recommendations: ✓ Implement pre-emptive evacuation by the City Disaster Risk Reduction Management Office (CDRRMO) and Barangay Disaster Risk Reduction Management Office (BDRRMO) during monsoon (heavy rainfall) and typhoon seasons. ✓ Construction and/or improvement of drainage system. Existing channels should be broadened and deepened to ensure smooth drainage of floodwaters/runoff. Avoid improper waste disposal on channels as it may cause clogging and obstruction of water flow. Design of canals should also consider obstructions by buildings/infrastructures. ✓ Strengthen and improve flood warning system with corresponding warning levels for pre-emptive evacuation purposes ✓ Adhere to the warnings given by proper authorities prior to the landfall or passing of typhoon and other prevailing weather disturbances. This may include CDRRMO, Office of the Civil Defense, and other related agencies. ✓ Construction of appropriate and well-studied engineering measures (e.g. dike or flood wall etc.) should be implemented to prevent overtopping of floodwaters and the fast rate of erosion along rivers/creeks. Make sure that the structure follows engineering standards and has structural integrity. However, even though there are engineering measures, communities near the river/creeks must remain vigilant and not complacent during strong typhoons and inclement weather. ✓ The barangay should have constant communication with nearby/upper barangays during typhoon and monsoon season. This is to gather information from their counterparts in terms of the flood situation in their areas. Usually, floodwaters coming from nearby/upper barangays are the main cause of flooding within the barangay area. This may serve as preparation for evacuation of their constituents, especially those near rivers/creeks/canals. ✓ The CDRRMO, in coordination with the BDRRMO, should conduct information and education campaigns on flood hazards to increase public awareness, especially to identified flood-prone communities.			

Purok / Sitio / Area of Concern	GPS Coordinates (Coordinate System: Luzon 1911)	Landslide Susceptibility Rating	Geohazards Assessment
Balete Road	14°39'06.9" N 121°07'26.8" E	High	Isolated steep slopes/ridges as a result of urbanization and/or human intervention may be susceptible to mass wasting particularly rock fall. Such situations can also be found in Brgy. Fortune. City and barangay officials reported previous landslide in a residential area along Balete Road. The area is relatively steep with several houses established along the slope, possibly overloading the slope.

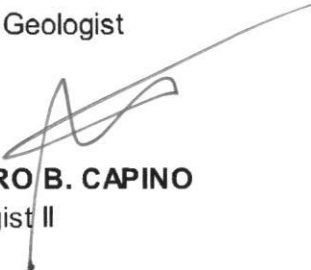
Remarks / Recommendations:

- ✓ **Observe presence of mass movement** (e.g. landslides, tension cracks) within the identified critical area/steep slopes and report to the local authorities and/or the MGB.
- ✓ **Observe for saturated ground and/or displaced roads surfaces** and report to the MGB/local authorities.
- ✓ **Implementation and maintenance of appropriate slope stabilization** (or slope protection/engineering intervention) measures along identified critical areas and roadcuts are highly recommended.
- ✓ Avoid settlement along the footslope, edge of the cliff/ravine, steep slope and ridge of hill/mountain.
- ✓ Barangay Disaster Risk Reduction Management Team must **implement monitoring and pre-emptive evacuation** during monsoon and typhoon season where extreme heavy rainfall occurs.
- ✓ **Activate** Barangay Disaster Risk Reduction Management Team in cooperation with the City Disaster Risk Reduction Management Team during monsoon and typhoon season.
- ✓ The CDRRMO together with the BDRRMO should conduct information and education awareness on this particular type of geohazard, particularly on landslide hazard to the landslide-prone communities.

Prepared by:


JULIUS VINCENT UMALI
Senior Geologist

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Senior Geologist

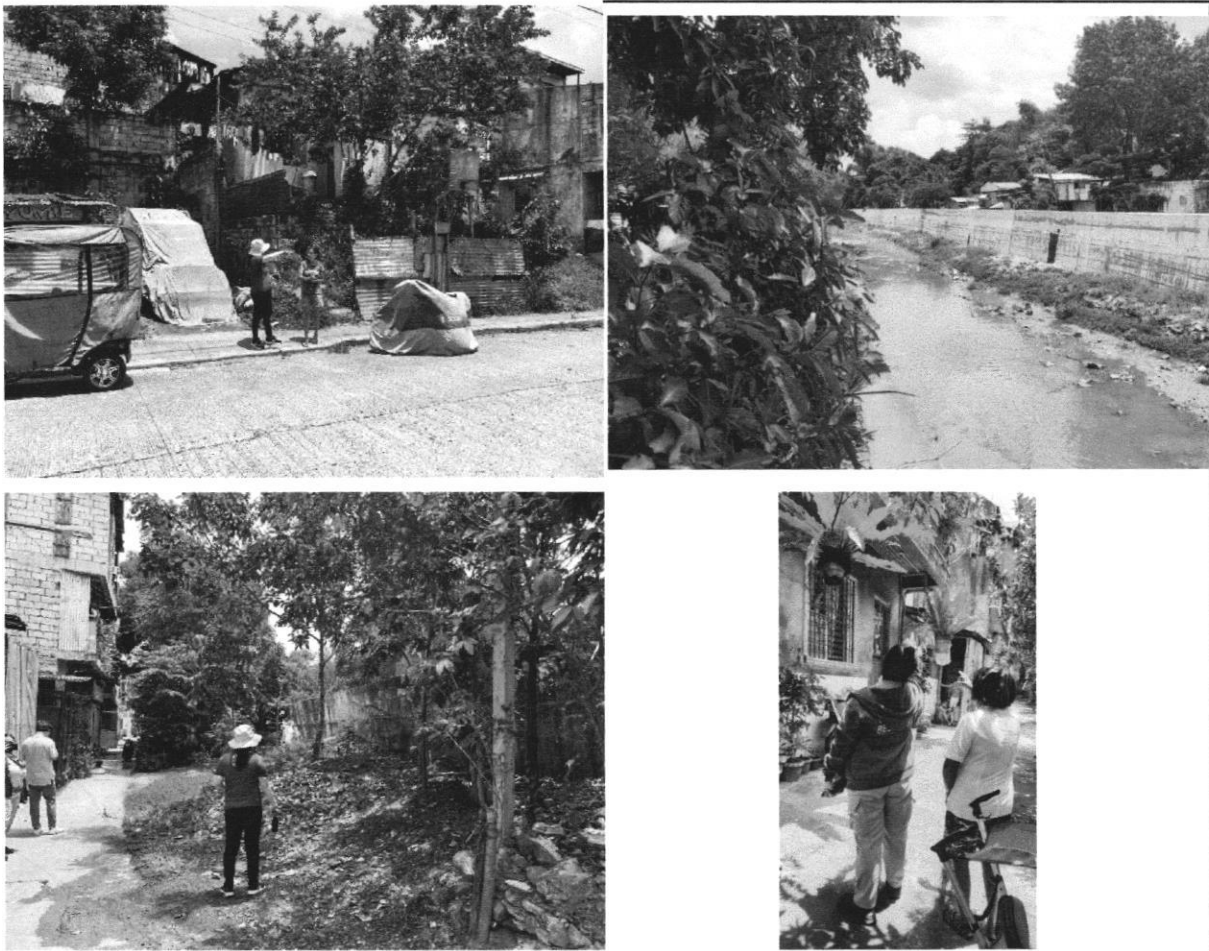

JETHRO B. CAPINO
Geologist II

Received by:

Designation:

Office:

PHOTOS



The overtopping of the channel during inclement weather conditions has been a major source of flooding within the barangay. Flood heights of more than 2 m were reported on residential settlements close to the said channel such as in San Miguel Phase II and III. Further, evidences of riverbank erosion were reported and observed especially on roads adjacent to the channel and/or its tributaries.

Recommendations:

- ✓ **Implement pre-emptive evacuation** by the City Disaster Risk Reduction Management Office (CDRRMO) and Barangay Disaster Risk Reduction Management Office (BDRRMO) during monsoon (heavy rainfall) and typhoon seasons.

- ✓ **Construction and/or improvement of drainage system.** Existing channels should be broadened and deepened to ensure smooth drainage of floodwaters/runoff. Avoid improper waste disposal on channels as it may cause clogging and obstruction of water flow. Design of canals should also consider obstructions by buildings/infrastructures.
- ✓ **Strengthen and improve** flood warning system with corresponding warning levels for pre-emptive evacuation purposes
- ✓ **Construction of appropriate and well-studied engineering measures** (e.g. dike or flood wall etc.) should be implemented to prevent overtopping of floodwaters and the fast rate of erosion along rivers/creeks. Make sure that the structure follows engineering standards and has structural integrity. However, even though there are engineering measures, communities near the river/creeks must remain vigilant and not complacent during strong typhoons and inclement weather.
- ✓ Adhere to the warnings given by proper authorities prior to the landfall or passing of typhoon and other prevailing weather disturbances. This may include CDRRMO, Office of the Civil Defense, and other related agencies.
- ✓ The barangay should have **constant communication with nearby/upper barangays** during typhoon and monsoon season. This is to gather information from their counterparts in terms of the flood situation in their areas. Usually, floodwaters coming from nearby/upper barangays are the main cause of flooding within the barangay area. This may serve as preparation for evacuation of their constituents, especially those near rivers/creeks/canals.
- ✓ The CDRRMO, in coordination with the BDRRMO, should conduct information and education campaigns on flood hazards to increase public awareness, especially to identified flood-prone communities.



City and barangay officials reported previous landslide in a residential area along Balete Road. The area is relatively steep with several houses established along the slope, possibly overloading the slope.

Recommendations:

- ✓ **Observe presence of mass movement** (e.g. landslides, tension cracks) within the identified critical area/steep slopes and report to the local authorities and/or the MGB.
- ✓ **Observe for saturated ground and/or displaced roads surfaces** and report to the MGB/local authorities.
- ✓ **Maintenance and/or implementation of appropriate slope stabilization** (or slope protection/engineering intervention) measures along identified critical areas and roadcuts are highly recommended.
- ✓ Barangay Disaster Risk Reduction Management Team must **implement monitoring and pre-emptive evacuation** during monsoon and typhoon season where extreme heavy rainfall occurs.
- ✓ **Activate** Barangay Disaster Risk Reduction Management Team in cooperation with the City Disaster Risk Reduction Management Team during monsoon and typhoon season.
- ✓ The CDRRMO together with the BDRRMO should conduct information and education awareness on this particular type of geohazard, particularly on landslide hazard to the landslide-prone communities.